

Emulation of SPHEREx Galaxy Power Spectra I: Neural Network Details and Optimization

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We present neural networks to generate redshift-space galaxy power spectrum multipoles for multiple tracer and redshift bins simultaneously given a set of input cosmology and galaxy bias parameters. This emulator utilizes a combination of traditional fully-connected network layers and transformer architecture to accurately predict galaxy power spectrum multipoles 680 times faster than traditional EFT calculations. We quantify network performance using both $\Delta\chi^2$, and likelihood contours for simulated SPHEREx analyses, using two correlated tracer bins and two independent redshift bins. After optimizing network architecture, the loss function, and training set sampling strategy, our emulator achieves $\text{median}(\Delta\chi^2) = 0.088$, and shows excellent agreement with EFT predictions at the contour level over a realistic parameter range. This work lays the foundation for future paper that will apply our emulator to run simulated analyses studying the impact of the covariance matrix, and represents an important step toward fast, accurate impact studies for the SPHEREx mission.

I. INTRODUCTION

Cosmologists have access to an ever-increasing library of data to work with. Currently, stage III missions like the Dark Energy Survey (DES [1]), the Hyper-Suprime Camera (HSC, [2]), and the Atacama Cosmology Telescope (ACT, [3]) are at or near completion, while several stage IV missions like the Dark Energy Spectroscopic Instrument (DESI, [4]), and Euclid [5] have already begun releasing early results. This trend will only continue with new surveys coming online, such as Rubin Observatory’s Legacy Survey of Space and Time¹ (LSST, [6]), and the Nancy Grace Roman Space Telescope².

One such new mission is the Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer (SPHEREx), which launched in early March 2025. SPHEREx is probing the large-scale structure (LSS) of the late universe by conducting an all-sky near-infrared ($0.75 < \lambda < 5 \mu m$) spectroscopic galaxy survey [7]. Over the course of two years, we expect to receive precise redshift measurements for ~ 450 million galaxies, which will allow us to precisely estimate the galaxy power spectrum and bispectrum at very large scales [8].

Analyzing these summary statistics will place tight constraints on early-universe inflation by measuring imprints of local-type primordial non-Gaussianity (PNG) on the late-time matter distribution through the parameter f_{NL}^{loc} (henceforth refereed to as simply f_{NL}). This

parameter is typically ~ 0.01 in single-field inflation models and > 1 in multi-field theories [9, 10]. Currently, the tightest constraints from CMB data (e.g., Planck) have $\sigma f_{NL} \sim 10$ [11] and are mainly limited by cosmic variance. Recent full-shape analyses using BOSS galaxies [12], eBOSS quasars [13, 14], and the DESI LRG sample [15] have found $\sigma(f_{NL}) \in [20, 40]$. These galaxy clustering measurements, while currently less precise than CMB constraints, can be significantly improved by increasing the effective survey volume. Indeed, SPHEREx aims to measure f_{NL} with a sensitivity of $\sigma(f_{NL}) \sim 1$ as its primary cosmological science objective, allowing us to differentiate between the two inflation scenarios [16].

To prepare for constraining f_{NL} with SPHEREx, specifically with the power spectrum, we need to quantify how specific analysis choices impact the resulting constraints via simulated likelihood analyses. These choices include investigating the specific priors used for both cosmology and galaxy bias parameters, what scales to include in the final analysis, and the impact of the covariance matrix, to name a few. However, performing these analyses involves predicting the galaxy power spectrum at $\approx 10^6$ different points of parameter space to adequately sample the posterior distribution. Most modeling codes take $\approx 1s$ per evaluation, which presents a significant computational barrier for this approach.

To enable fast evaluations during likelihood analyses, we present a fast method for emulating galaxy power spectrum multipoles using machine learning. We develop a set of deep neural networks to directly emulate the galaxy power spectrum for multiple tracer and redshift bins at once by using a combination of fully connected layers and modern transformer architecture [17]. Our emulators are trained on SPHEREx-like galaxy power spectrum multipoles predicted by Eulerian effective-field theory (EFT, [18]). We optimize various network hyperparameters, as well as the training set sampling strategy,

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to maximize accuracy. We then use our emulator to run SPHEREx-like analyses within Λ CDM using two correlated tracer bins, and two independent redshift bins.

This paper is the first of two manuscripts presenting this galaxy power spectrum emulator. In our second paper, we will use our emulator to investigate various aspects of the covariance matrix w.r.t. the resulting parameter constraints, including the impact of non-Gaussian terms, the window function, and input cosmology. Here, we proceed as follows. Section II describes the galaxy power spectrum model we emulate, as well as how we generate a training set with said model. Section III discusses the neural network architecture and training scheme, while section IV discusses both how the emulator is optimized and the performance metrics we use to quantify accuracy. Finally, we use our emulator in a simulated SPHEREx likelihood analysis in Section V, and summarize our findings in Section VI.

II. EMULATING THE GALAXY POWER SPECTRA

A. EFT Model Details

We emulate the Eulerian effective field theory (EFT) prediction for the redshift-space galaxy power spectrum. This model has been used extensively to analyze BOSS data with both Gaussian initial conditions [18–21], and local PNG [12]. More recently, it was also used to independently constraint dark energy models with DESI data [22]. To summarize, this EFT model splits the full galaxy power spectrum into four components (omitting IR resummation and the Alcock-Paczynski effect in our expressions for simplicity),

$$P^{\text{EFT}}(k, \mu) = (b_1 + f\mu^2)P_{\text{lin}}(k) + P_{1\text{-loop}}(k, \mu) + P_{\text{ctr}}(k, \mu) + P_{\text{shot}}(k, \mu), \quad (1)$$

where f is the growth factor at the effective redshift of the given galaxy sample, μ is the cosine line-of-sight angle, and b_1 is linear galaxy bias. The first term in Eq. 1 describes the linear Kaiser effect [23], while the one-loop term corrects for non-linear structure growth and redshift-space distortions (RSD) effects. To calculate this term, we expand the galaxy density contrast³ $\delta_g(\mathbf{x})$ with the following basis,

$$\delta_g(\mathbf{x}) = b_1\delta(\mathbf{x}) + \frac{b_2}{2}\delta(\mathbf{x})^2 + b_{\mathcal{G}_2}\mathcal{G}_2(\mathbf{x}) \quad (2)$$

where $\delta(\mathbf{x})$ is the nonlinear matter density contrast, $\mathcal{G}_2$ is the tidal field operator in Fourier space, and b_2 and $b_{\mathcal{G}_2}$ are second-order bias parameters. Following previous analyses of BOSS data, we set all third-order bias terms to 0 for this work. Next, $P_{\text{ctr}}(k, \mu)$ corrects for small-scale physics integrated over by the one-loop term [24]. Following the literature, we include two multipole-specific terms and one next-to-leading order term with the following form,

$$P_{\text{ctr}}(k, \mu) = -2c_0k^2P_{\text{lin}}(k) - 2c_2k^2P_{\text{lin}}(k) - \tilde{c}f^4k^4(b_1 + f\mu^2)P_{\text{lin}}(k), \quad (3)$$

where $c_0, c_2, \tilde{c}$ are free parameters. This term is cheap to calculate provided one has access to the linear power spectrum. Thus, we do not directly emulate $P_{\text{ctr}}(k, \mu)$, and instead calculate it normally after emulating the tree and one-loop terms.

Finally, P_{shot} accounts for stochastic contributions to the power spectrum. Similar to the counterterms, we do not emulate this term directly due to it being trivial to calculate. In total, the Eulerian EFT terms contains three emulated nuisance parameters ($b_1, b_2, b_{\mathcal{G}_2}$), and four analytically calculated parameters ($c_0, c_2, \tilde{c}, P_{\text{shot}}$). When adding IR resummation and the Alcock-Paczynski effect, this model produces accurate predictions up to $k = 0.3$ h/Mpc.

B. Generating a Training Set

We generate training sets based on the EFT model described above. Specifically, we compute four million sets of (tree and one-loop term only) galaxy power spectrum $l = (0, 2)$ multipoles using the upcoming EFT code `ps.1loop` [25] in an embarrassingly-parallel fashion. This code is similar to *CLASS-PT* [18], but also accounts for local PNG using the formalism in Ref. [12]. For each set of input cosmology and bias parameters, we output multipoles for two independent redshift bins ($0.2 < z < 0.4, 0.4 < z < 0.6$), and two correlated tracer bins. The specific number densities and redshift ranges were chosen to mirror those expected from SPHEREx. We opt to use 25 linearly-space k -bins with $k_{\text{max}} = 0.2$ h/Mpc.

We generate a training set with the above configurations by sampling parameters uniformly within a hypersphere [26]. Hypersphere sampling effectively cuts out low-probability regions of parameter space, while simultaneously increasing the density of points within the range of interest compared to a Latin hypercube. We investigate this more novel sampling strategy later by also generating a training set with a Latin hypercube. In both cases, we use the parameter ranges listed in Table I. For actual training, we use 80% of the full dataset for updating network weights during training, 10% for validation during training, and 10% for evaluating the final model performance.

³ In this paper, we assume Gaussian initial conditions. When accounting for local PNG, this expansion includes several more bias terms that depend on f_{NL} .

To facilitate the emulator training process, we normalize all input parameters to lie within the range $[0, 1]$. Additionally, we further post-process our training set in a similar fashion as Refs. [27, 28]. Given a data covariance matrix, we first compute its eigenvalue decomposition,

$$C = Q\Sigma Q^{-1}. \quad (4)$$

We obtain C via a modified version of CovaPT [29] configured to output multi-tracer Gaussian covariance matrices. Next, we transform each set of power spectrum multipoles to lie within the diagonal basis defined by the eigenvectors Q using the following equation,

$$\hat{P}_i = \frac{Q^{-1}P_i - Q^{-1}P_{fid}}{\sqrt{\Sigma}}. \quad (5)$$

Here, P_{fid} are the galaxy power spectrum monopole and quadrupole calculated using the fiducial cosmology in Table I. Our emulator is thus trained to output $\hat{P}$, which is then post-processed back into galaxy power spectrum multipoles.

III. NEURAL NETWORK ARCHITECTURE

Our emulator utilizes a modular design, in which we assign independent networks to each set of (auto or cross)

Parameter	Fiducial Value	Emulator Range
h	0.6736	$U[0.44, 0.9]$
ω_{cdm}	0.1201	$U[0.0701, 0.1701]$
ω_b	0.02218	$U[0.01998, 0.02438]$
A_s	2.1×10^{-9}	$U[1.2 \times 10^{-9}, 3 \times 10^{-9}]$
n_s	0.96589	$U[0.83989, 1.09189]$
$b_1^{(0,0)}$	1.5	$U[0.25, 2.75]$
$b_1^{(0,1)}$	1.3	$U[0.05, 2.55]$
$b_1^{(1,0)}$	1.8	$U[0.55, 3.05]$
$b_1^{(1,1)}$	1.4	$U[0.15, 2.65]$
$b_2^{(0,0)}$	-1.37	$U[-3.12, 0.38]$
$b_2^{(0,1)}$	-1.57	$U[-3.32, 0.18]$
$b_2^{(1,0)}$	-0.78	$U[-2.53, 0.97]$
$b_2^{(1,1)}$	-1.49	$U[-3.24, 0.26]$
$b_{\mathcal{G}_2}^{(0,0)}$	-0.29	$U[-1.79, 1.21]$
$b_{\mathcal{G}_2}^{(0,1)}$	-0.17	$U[-1.67, 1.33]$
$b_{\mathcal{G}_2}^{(1,0)}$	-0.46	$U[-1.96, 1.04]$
$b_{\mathcal{G}_2}^{(1,1)}$	-0.23	$U[-1.73, 1.27]$

TABLE I. Table describing the fiducial cosmology and parameter bounds used in our training sets and simulated analyses. For cosmology parameters, we adapt the priors and fiducial values from the baseline analysis in Ref. [30]. For galaxy bias, we set different fiducial values for each tracer and redshift bin for b_1 , and estimate the fiducial values for b_2 and $b_{\mathcal{G}_2}$ using the fits from Refs. [31, 32]. We set all other bias parameters to 0 unless otherwise specified.

power spectrum multipoles corresponding to an individual tracer and redshift bin. Since we are correlating different tracers within a given redshift bin, we have three networks per redshift bin (two auto, one cross spectra) for our chosen configuration. This design choice allows us to keep each network relatively small, making the emulator easier to train and expand to larger tracer / redshift bin setups. Each network receives all cosmology and only the relevant galaxy bias parameters for that particular bin. We provide a schematic of our emulator in Fig. 1.

The networks themselves follow a similar design to the complete attention-based design from Ref. [28]. We start by passing the (normalized) cosmology and bias parameters through an initial input layer, then through a series of four fully-connected MLP blocks. Each of these blocks has two MLP layers with batch normalization, followed by a residual connection. The output of those blocks then gets sent to an embedding layer that doubles the dimensionality to one hundred, before passing through three transformer blocks. In each of these blocks, the input goes through a scaled dot-product attention mechanism, then split into ten equally-sized components that are each fed to separate feed-forward layers. Finally, the transformer block output passes through a final dense layer, which outputs the galaxy power spectrum monopole and quadrupole moments for a specific tracer and redshift bin combination. We chose the specific number of MLP and transformer blocks, as well as the number of layers per MLP block, based on the optimization results in Section IV A.

After each layer within the MLP blocks, and the feed-forward layer in the transformer blocks, we use the following nonlinear activation function,

$$h(x) = \left(\gamma + (1 + e^{-\beta \odot x})^{-1} \odot (1 - \gamma) \right) \odot x, \quad (6)$$

where β, γ are trainable parameters of the model with the same shape as the input vector x . This function is thought to more accurately capture both sharp and smooth features due to its flexibility, and has been used by other emulators in the field [28, 33, 34].

Each network is trained separately by stochastic gradient descent. We tested several different loss functions to minimize, all of which are based on the $\Delta\chi^2$ between the emulator output and the prediction from our EFT model,

$$\Delta\chi_{(z,tt)}^2 = \left(P_{(z,tt)}^{\text{emu}} - P_{(z,tt)}^{\text{EFT}} \right)^T C_{(z,tt)}^{-1} \left(P_{(z,tt)}^{\text{emu}} - P_{(z,tt)}^{\text{EFT}} \right). \quad (7)$$

Here, z denotes the specific redshift bin, tt describes the specific tracer combination (in our case, 11, 12, or 22), and C is the data covariance matrix. We describe the specific loss functions we tested in Section IV A.

We utilize the Adam optimization scheme for updating our network weights [35]. To allow for adequate loss function exploration without preventing the networks from

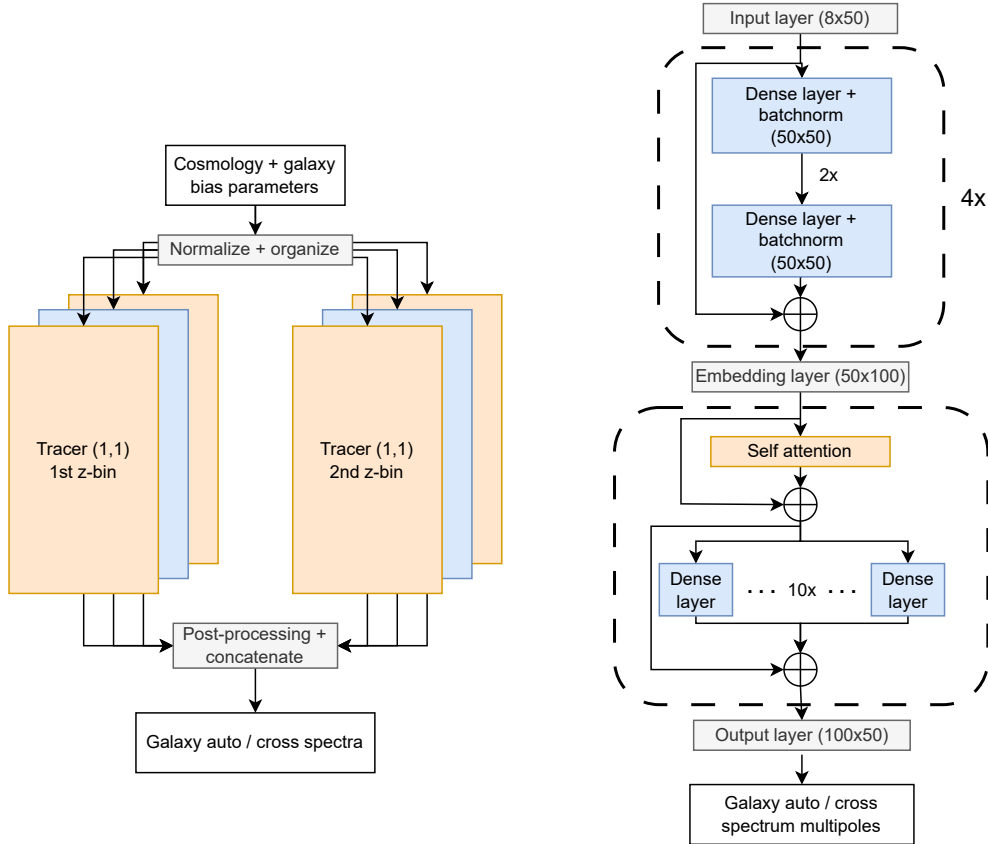


FIG. 1. (Left): Schematic of our full galaxy power spectrum emulator. Cosmology and galaxy bias parameters are normalized and passed to six independent neural networks, each responsible for generating the auto or cross power spectrum for a particular tracer and redshift bin. We then concatenate the output from these networks to output the galaxy power spectrum for two tracer and two redshift bins.

(Right): Architecture of an individual network in the full emulator. Relevant parameters first pass through a series of four MLP blocks, followed by an embedding layer. After that, we proceed through one transformer encoder block using the same architecture as Ref. [28], before finally outputting normalized multipoles for a specific redshift and tracer bin.

reaching a minimum, we also implement an adaptive learning rate scheduler that reduces the learning rate by a factor of 10 if the validation set loss does not improve after fifteen epochs. Finally, to prevent overfitting we use early stopping where training ends after the validation loss has not improved after twenty-five epochs.

IV. OPTIMIZATION AND TESTING

A. Hyperparameter Optimization

We optimize the emulators by manipulating several hyper-parameters, which each can have a large impact on model performance that is not known apriori. To quantify those impacts, we individually vary the initial learning rate, batch size, number of MLP blocks, number of layers per MLP block, and number of transformers

blocks. We define the list of values we iterate through in Table II. The default parameter values were chosen based on Ref. [28], which uses a similar network design to emulate LSST observables. We iteratively train emulators using these values, varying one hyperparameter at a time to isolate the effect each value has on the resulting performance. We use the median $\Delta\chi^2$ calculated from the test set using equation 7 as our main performance metric.

We also explore two different loss functions given by the following equations,

$$\mathcal{L}_1 = \langle \Delta\chi^2 \rangle, \quad (8)$$

$$\mathcal{L}_2 = \langle \sqrt{1 + 2\Delta\chi^2} \rangle - 1, \quad (9)$$

which we call χ^2 and hyperbolic χ^2 loss respectively. We expect $\mathcal{L}_1$ to be more sensitive to outliers than $\mathcal{L}_2$, as it depends on the difference between emulated and true power spectra squared. For each loss function, we iter-

Hyperparameter	Default Value	Iteration Values	Final Value
Learning Rate	0.001	0.0001, 0.0005, 0.0025, 0.005, 0.0075, 0.01	0.005
Batch Size	200	50, 100, 300, 500	200 ^a
MLP Blocks	2	1, 3, 4, 5	4
Layers / Block	4	2, 3, 5, 6	2
Transformer Blocks	1	2, 3, 4, 5	1 ^a

^a Set to non-optimal value after accounting for training time.

TABLE II. Table of network hyperparameters varied in our optimization scheme. We train emulators for each iteration value, holding all other hyperparameters at their default values. We use $\text{median}(\Delta\chi^2)$ to determine which configuration performs best, the results of which (after accounting for training time) are displayed in column 4.

atively vary the hyper-parameters in table II to account for potentially different optimal configurations.

Table II shows our final hyperparameter setup. Notably, batch size appears to have an inverse relationship with both performance and training time. Thus, we chose to adopt a final batch size of 200 to keep our results more easily scalable to larger setups. Similarly, increasing the number of transformers marginally improves performance, but also significantly increases training time. Thus, for the same reasons as above, we opt to use 1 block as our final value.

Additionally, we find that the above conclusions hold regardless of the loss function used. In fact, neither loss function performs consistently better or worse than the other, with hyperbolic χ^2 performing 0.5% better when averaging over all our training runs. These differences may be accounted for by the training process being inherently noisy, thus we conclude that both $\mathcal{L}_1$ or $\mathcal{L}_2$ perform the same for this problem. In what follows, we chose to use $\mathcal{L}_2$ as our "optimal" loss function.

B. Training Set Optimization

Next, we explore the impact of the training set by varying both the number of power spectra we use to train, and the method of sampling the input cosmology and bias parameters. Both tests are done using the optimal set of hyper-parameters found from our above efforts.

The top panel of Figure 2 shows $\Delta\chi^2_{tot}$ histograms for emulators trained on differently-sampled training sets. We see that using hypersphere sampling results in two orders of magnitude improvement in performance compared to using a traditional Latin hypercube. Such a stark improvement can likely be attributed to the overall increased density given the same number of samples [26].

Next, the bottom panel of Figure 2 shows how the median $\Delta\chi^2$ changes with the number of training set samples. We find that both sampling strategies roughly follow a power law with the form,

$$\text{median}(\Delta\chi^2) = A(N_{sample})^{-x}, \quad (10)$$

with $x = (0.798, 0.771)$ for hypersphere and hypercube

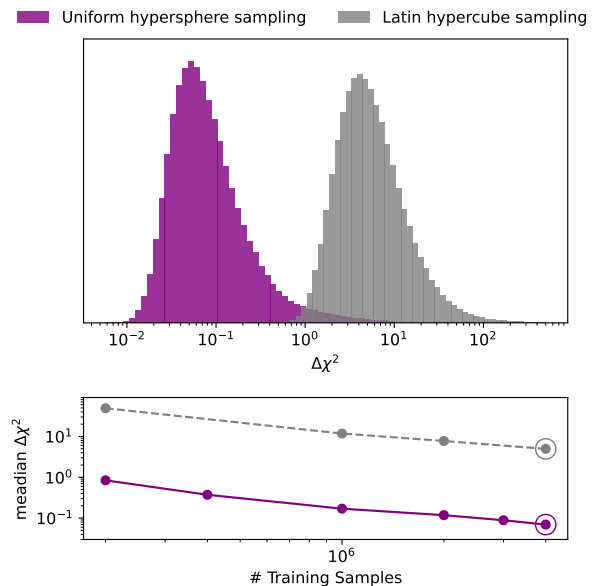


FIG. 2. (Top): Histogram of $\Delta\chi^2_{tot}$ values for emulators trained on training sets sampled with a Latin hypercube (gray) and a uniform hyper-sphere (purple). We see roughly two order of magnitude difference between the two sampling strategies.

(Bottom): Median $\Delta\chi^2$ with respect to the total number of training samples for each sampling strategy. Circled points correspond to the histograms in the top panel. In both cases, performance improves roughly following a power law after $\sim 10^6$ samples.

sampling respectively. Extrapolating outwards, we can naively expect performance to double after increasing the training set size by a factor of around 2.4 in both cases. This result follows other scaling relations seen in a broad range of machine learning problems [36, 37]. Here, this relation means that our model could still achieve greater accuracy with larger training sets, which may become important when increasing the number of tracer and redshift bins.

Finally, we present more detailed $\Delta\chi^2$ statistics for our final optimized emulator. Figure 3 displays $\Delta\chi^2$ his-

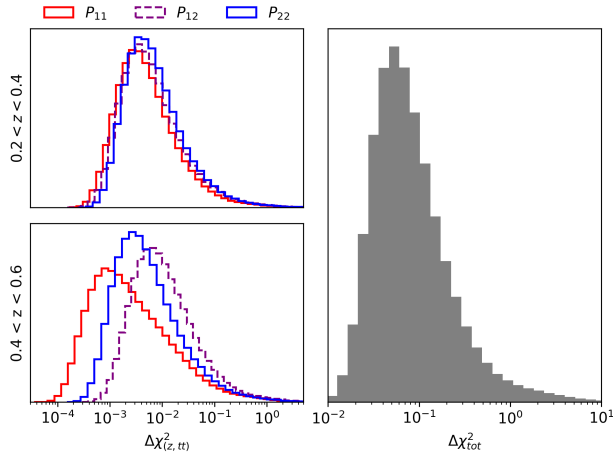


FIG. 3. (Left): Histograms of $\Delta\chi^2$ for each individual network in the final optimized emulator. Each panel denotes a specific redshift bin, and we denote auto/cross power spectra with solid and dashed lines respectively. (Right): Histogram of the total $\Delta\chi^2$ calculated from the full optimized emulator output.

tograms for each sub-network, and also for the full emulator output. We see that each network performs quite well in this metric, with most histograms peaking around 10^{-2} . The higher redshift bin show notably larger variation between networks, with P_{11} peaking at an impressive 10^{-3} . Looking at the full output, we see a slightly higher distribution with a median of 0.069. This increase is actually larger than simply summing all of the sub-network errors, due to including off-diagonal blocks of the covariance matrix that aren't accounted for when calculating $\Delta\chi^2_{(z,tt)}$. As a consequence, we expect that increasing the number of tracer and redshift bins will result in larger $\Delta\chi^2_{tot}$, even if each sub-network's accuracy is kept the same as here.

Figure 4 shows the median $\Delta\chi^2_{tot}$ over the emulated parameter space. We can see that performance is relatively constant, with larger variation in some pixels along the edges. This trend indicates that we should not see any bias due to variations in performance when running simulated analyses, so long as said analyses lies close to the center of each range.

V. COMPARING EMULATED & BRUTE FORCE SPHEREX SIMULATED LIKELIHOOD ANALYSES

A. Metric 1: Parameter Contours

We run simulated analyses to showcase our optimized emulator by varying all cosmology and bias parameters listed in Table I, assuming flat curvature and no PNG ($f_{NL} = 0$). Fiducial values for cosmology parameters are taken from the baseline DESI Y1 full modeling analyses

[30]. For galaxy bias, we assume different values of b_1 for each tracer and redshift bin, and calculate fiducial values for $b_2, b_{\mathcal{G}_\epsilon}$ using fits derived in Refs. [31, 32]. We use all flat priors corresponding to our emulator ranges, except for ω_b and n_s , which use Gaussian priors taken from the same baseline DESI-Y1 analysis,

$$\omega_b \sim \mathcal{N}(0.02218, 0.00055^2), n_s \sim \mathcal{N}(0.96589, 0.042^2). \quad (11)$$

Since our emulator uses explicit ranges for these parameters, we also apply a cutoff with the ranges given in Table I, corresponding to $\pm 4\sigma, \pm 3\sigma$ respectively. Finally, we neglect the effect of massive neutrinos for these analyses, so $m_\nu = 0$ eV.

In addition to the above parameters, we additionally vary $c_0, c_2, \tilde{c}$, and P_{shot} , by including those power spectrum terms analytically with the following priors,

$$c_0, c_2 \sim \mathcal{N}(0, 30^2), \quad \tilde{c} \sim \mathcal{N}(500, 500^2), \quad (12)$$

$$P_{shot} \sim \mathcal{N}(2000, 5000^2). \quad (13)$$

These ranges are taken from Ref [20], with the additional change that the stochastic parameter is instead centered on $1/n_g$ of each tracer / redshift bin. We use `symbolic.pofk`⁴ [38] to quickly calculate the linear power spectrum required for calculating the counterterms.

For all analyses, we assume a Gaussian likelihood, with the same Gaussian covariance matrix used to train our emulator. We present the resulting contours in terms of $[H_0, \Omega_m, \sigma_8]$ as those constraints are more prevalent in the literature. Our chains are run using the *Nautilus* python package [39], which estimates both the posterior distribution and evidence factor $\log \mathcal{Z}$ using importance-nested sampling. Finally, we generate our contour plots using *GetDist* [40].

Our first analysis includes only a single tracer and redshift bin from our full setup. We chose the higher redshift bin ($0.4 < z < 0.6$), and the lower redshift-uncertainty tracer bin. Overall, this setup has a total of twelve varied parameters.

Figure 5 shows the marginalized posteriors from this first analysis, while Table III presents the 1D bestfit shifts for the cosmology parameters between our emulator and *ps_1loop* chains. By eye, we see good agreement between emulated and analytic contours. More quantitatively, the resulting 1D marginalized bestfit values have an average shift of 0.057σ , with h having the largest deviation at 0.14σ . We also see the 1D error bars agree nicely with an average deviation of 2.2%, and P_{shot} having the highest deviation at 5.1%.

Next, we run analyses using all tracer and redshift bins in our setup. Since each bin has independent nuisance

⁴ https://github.com/DeaglanBartlett/symbolic_pofk

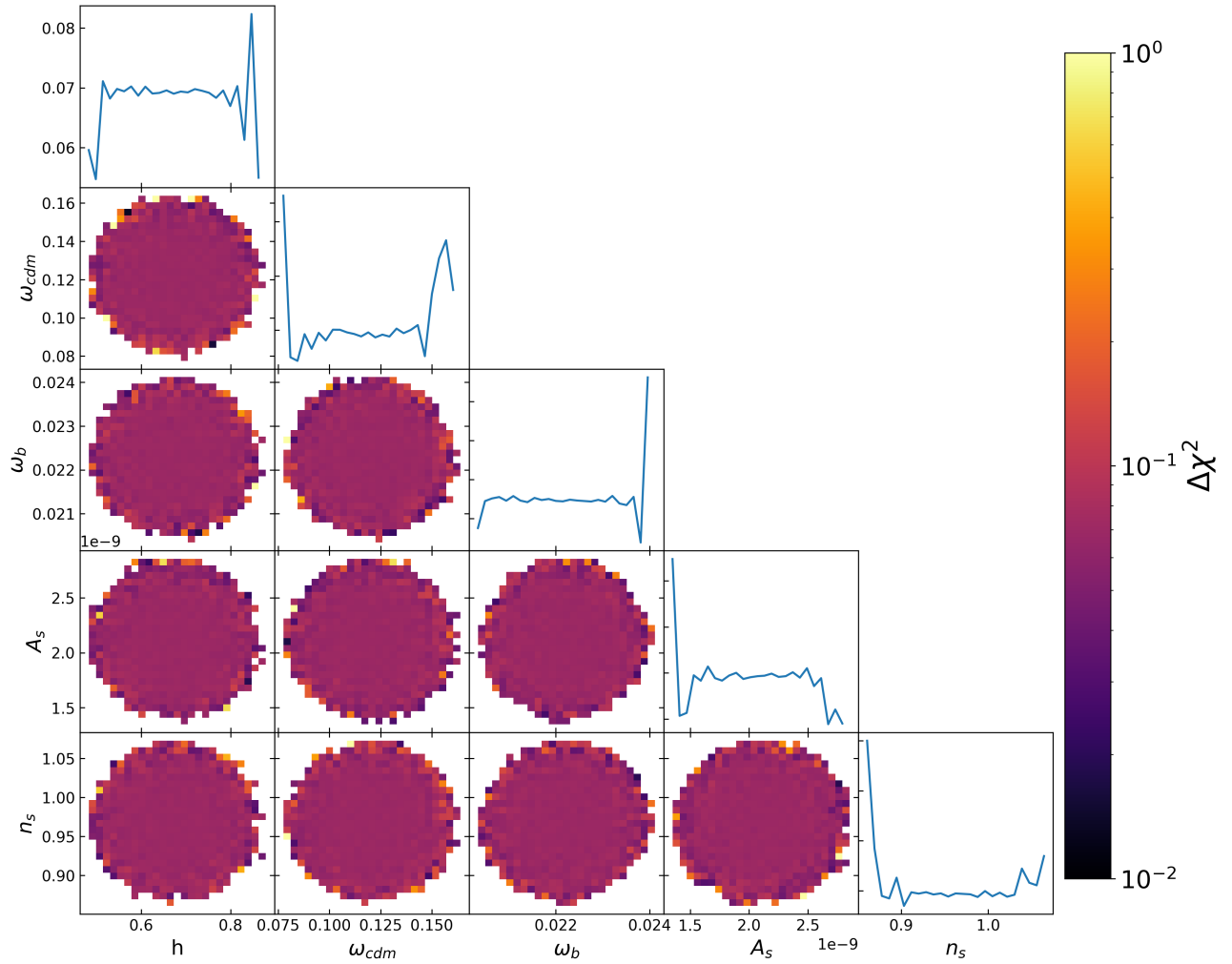


FIG. 4. Heatmap showing the median $\Delta\chi^2$ of our emulator throughout parameter space. Each pixel represents the median value from samples in the test set whose parameters fall in that specific range. Other than some regions along the edges, the emulator performs similarly across the entirety of parameter space.

Parameter	1t_1z		2t_2z	
	best-fit shift	errorbar shift	best-fit shift	errorbar shift
$\Delta \log \mathcal{Z}$	1.7×10^{-2} (0.062%)		7.8×10^{-1} (0.90%)	
h	0.143σ	0.73%	0.16σ	2.33%
ω_b	0.094σ	1.63%	0.21σ	1.96%
ω_{cdm}	0.096σ	1.21%	0.14σ	1.05%
A_s	0.071σ	2.70%	0.12σ	1.01%
n_s	0.051σ	0.64%	0.10σ	0.951%
Ω_m	0.031σ	0.92%	0.019σ	0.034%
σ_8	0.024σ	0.02%	0.072σ	0.855%

TABLE III. Shift of 1D best-fit values and shift in 1D errorbars for varied cosmology parameters when comparing chains produced with our emulator vs *ps_1loop*. The values in column 2 and 3 correspond to Figure 5, and columns 4 and 5 to Figure 6. We also display the change in Bayesian evidence between emulated and analytical analyses for each configuration.

parameters, we now have a total of thirty three varied parameters.

JOE: Numbers in the below paragraph + Figure 6 still need to be updated Figure 5 shows the marginalized posteriors from this multi-tracer analysis,

while Table III presents the 1D bestfit shifts. We again see good agreement between the emulated and analytic chains, with all best-fit values agreeing to within 0.21σ and error bars to within 2.3%. These shifts are somewhat higher than the single tracer and redshift bin anal-

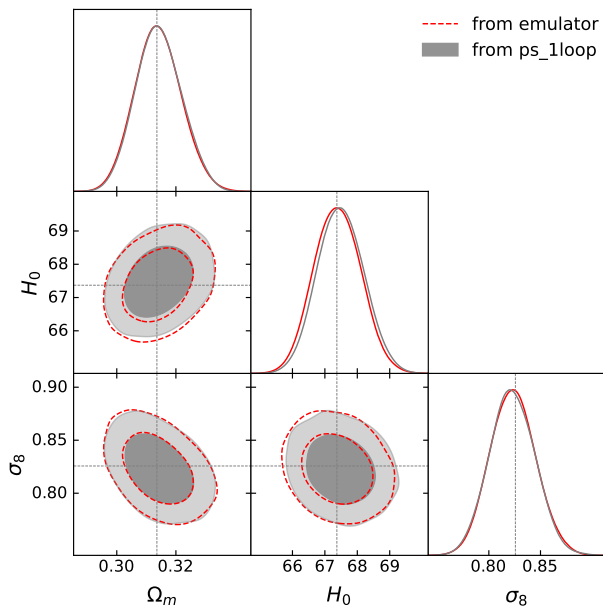


FIG. 5. Parameter contours from simulated likelihood analyses with one tracer and one redshift bin, using pure EFT calculations (gray) and our power spectrum emulator (red). Both contours lie almost directly on top of each-other, indicating our emulator broadly recovers the expected posterior distribution.

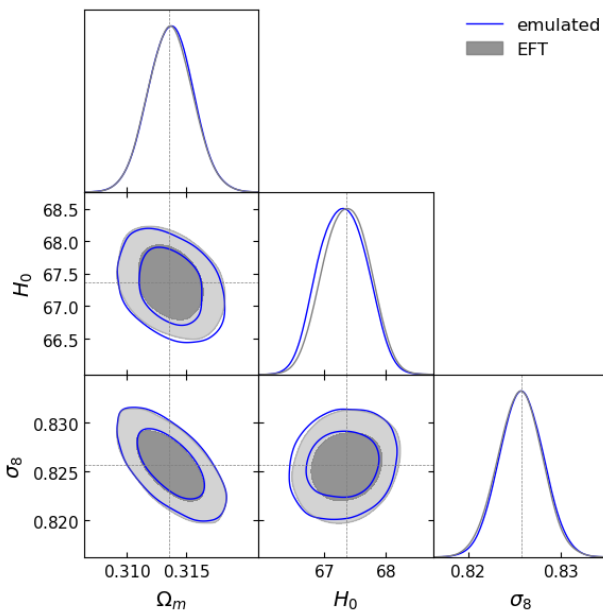


FIG. 6. **JOE: Waiting on chains to finish to replace** Same as Fig. 5, except with analyzing all tracer and redshift bins in our setup. Emulated and analytic contours show good agreement, with slightly larger deviations than our simpler analysis setup.

ysis, which mirrors the larger total $\Delta\chi^2$ in Figure 3 and suggests adding more tracer / redshift bins will result in larger deviations when keeping each sub-network performance constant. Even with this trend however, our emulator can broadly reproduce the expected posterior distribution accurate enough to run impact studies.

B. Metric 2: Bayesian Evidence

Finally, we report the change in Bayesian evidence factor $\Delta \log \mathcal{Z}$ between emulated and analytical chains for the above analyses. This quantity is frequently used to compare the goodness of fit of different models and datasets [41, 42], meaning we want as small $\Delta \log \mathcal{Z}$ as possible if we wish to use our emulator to study the impact of analysis choices on parameter constraints.

Table III displays $\Delta \log \mathcal{Z}$ for both tracer and redshift bin setups. In both cases, the difference is much smaller than the true evidence (0.062% and 0.90% respectively), indicating our emulator is able to give an accurate sense of the true model's goodness of fit. We also see the same trend as before, with the larger setup deviating slightly more than when using a single tracer and redshift bin.

VI. SUMMARY & CONCLUSION

Predicting the galaxy power spectrum serves as the main computational bottleneck when running simulated analyses for stage-IV surveys like SPHEREx. To enable faster impact studies in preparation for real data, this paper presented a fast method of generating galaxy power spectrum multipoles using a series of MLP and transformer-based neural networks. We used this neural network emulator to output the monopole and quadrupole moments predicted by the EFTofLSS for two correlated tracer bins, and two independent redshift bins, all simultaneously. In this setup, our emulator can output power spectrum multipoles in $\sim 3\text{ms}$ (0.5ms for one tracer / redshift bin combination) on a typical laptop, which is roughly 680 times faster than traditional EFT codes ran on the same hardware. The code used to train and use this emulator is publicly available on GitHub ⁵.

We optimized our emulator setup by individually varying several model and training hyperparameters, and using the median $\Delta\chi^2$ to determine how each one affected performance. We also tested two different loss functions during this process, and found that both achieved the same overall accuracy. Next, we explored how performance changes when changing both the sampling strategy used to generate the training set, and the number of training set samples. Overall, our emulator performance follows a clear power law w.r.t. training set size,

⁵ https://github.com/jadamo/spherex_emu

and uniform hypersphere sampling performs ~ 2 orders of magnitude better than using a traditional Latin hypercube. These results indicate we can further improve performance if needed by increasing the amount of training data,

We tested the accuracy of our optimized emulator by computing the $\Delta\chi^2$ error between emulated and analytically-calculated power spectrum multipoles in the test set for each individual tracer / redshift bin, and for all bins together. All sub-networks perform well, with median $\Delta\chi^2_{z,tt}$ all at or below 10^{-2} , and one bin achieving an impressive 10^{-3} . The full emulator output achieves a median $\Delta\chi^2_{tot} = 0.069$, with no observed parameter dependence. These results pass the target threshold $\Delta\chi^2 < 1$ used by the DES collaboration, and other emulators of cosmological observables [27, 43].

Finally, we ran simulated likelihood analyses using our emulator, varying five cosmology, three galaxy bias, one shot-noise, and three counterterm parameters. We ran analyses for both a single tracer and redshift bin, and for two of each, both using the same emulator. The resulting contours for the first analysis match with the expected distribution to great precision, with best-fit values agreeing to within 0.02σ , error bars to within 2.5%, and Bayesian evidence agreeing to 0.062%. Analyzing all bins resulted in slightly larger, but still small deviations, with best-fit values agreeing to 0.21σ , error bars to within 2.3%, and Bayesian evidence agreeing to 0.9%. Put together, these results indicate our emulator can reproduce the galaxy power spectrum accurate enough to run parameter impact studies for missions like SPHEREx.

This work corroborates recent efforts to emulate cosmological observables from both galaxy clustering and

weak-lensing surveys (e.g. Refs. [27, 28, 44]). In particular, we reach a similar conclusion as Ref. [44], who developed a pca-based neural network to predict LPT-based galaxy power spectra in a similarly sized parameter space. Given the similar use-cases, our emulator differs from theirs in the specific bias model used, the underlying network architecture, and the ability to concurrently generate power spectra for multiple tracer and redshift bins.

To conclude, our emulator can demonstrably generate galaxy power spectra in a large enough input parameter space to run simulated analyses for SPHEREx. Indeed, this manuscript is the first in a series, with the next entry investigating the impact of the data covariance matrix on constraining power using the infrastructure built here. Overall, we consider this work as both a useful tool towards a precise measurement of f_{NL} with SPHEREx, and as a further demonstration of the promise of using machine learning to enhance simulated likelihood analyses.

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